

GENETIC ENGINEERING

Engineering the Structural Logic of Genomic Modification & Bio-Design

A project-first certification designed to master CRISPR-Cas9, Viral Vectorology, and the Governance of Transgenic Systems. Architect the future of biological function.



THE EDLANX METHODOLOGY

OUR ENGINEERING PHILOSOPHY

At Edlanx, we don't just "edit genes"; we cultivate a Genomic Architect's mindset. Our program is a deep dive into the site-specific logic and structural stability of modified organisms. We believe in Design Integrity — moving beyond basic PCR to build a robust understanding of knock-out/knock-in strategies, metabolic engineering, and the rigorous regulatory governance required for biosafety.

THE FOUR PILLARS OF MASTERY

-  **Sequence Logic** Master the structural identification of target loci and PAM sequences.
-  **Modification Governance** Learn the logic of NHEJ and HDR repair pathways.
-  **Vector Architecture** Master the structural lifecycle of Plasmid, AAV, and Lentiviral delivery systems.
-  **Analytical Strategy** Bring designs to life by mastering the structural link between genotype modification and phenotype expression.

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WHY CHOOSE EDLANX



01 Industry-Ready Curriculum

We move beyond basic molecular biology to bridge the gap between theoretical genetics and industrial-scale bio-design. Our Genetic Engineering syllabus is a deep immersion into CRISPR-Cas9, Viral Vectorology, and Metabolic Engineering — the exact technical pillars used by biotech giants and clinical research organizations to architect the future of medicine.

02 Real-Time Corporate Exposure

Step into a high-fidelity technical environment at our Bangalore Operational Center. At Edlanx, you won't just study "DNA sequences"; you will be treated as a Genomic Architect, working in simulated laboratory environments to master site-specific editing, navigate bio-containment protocols, and manage the structural integrity of modified organisms.

03 Mentorship & Continuous Assessment

Your growth is guided by senior Geneticists and Bio-Process Engineers through structured mentoring and performance evaluations. We focus on developing your ability to design high-fidelity guide RNAs, troubleshoot off-target mutations, and translate raw sequence data into predictable, functional phenotypes.

04 Skill Mastery & Performance Incentives

Master industry-standard bioinformatics software and in-silico modeling tools while learning essential professional competencies. We believe in experiential learning and reward breakthroughs in transformation efficiency or metabolic yield with performance-based stipends, ensuring you remain focused on technical excellence.

05 Global Career Readiness

We enhance your employability by focusing on the Bio-Lifecycle, from initial target identification and vector design to clinical trial regulation and IP filing. Our career guidance includes mock "Genomic Sprints" and technical interviews tailored for top-tier roles in Gene Therapy, Agricultural Biotech, and Synthetic Biology startups.

06 Collaborative Engineering Community

Join a supportive culture built on transparency, technical integrity, and disciplined ethics. You will learn to lead a "bio-sprint," collaborate across computational and experimental domains, and grow your network within an elite community of researchers and genetic engineers.

07 Verified Professional Certifications

At Edlanx, we don't just issue certificates for attendance; we award them based on verified learning, real-world performance, and practical outcomes. Our credentials are proof of your industry-ready skills and are recognized by institutions and recruiters worldwide.

We provide four distinct levels of certification:

- Internship Completion Certificate: Validates your exposure to professional work culture, verified internship hours, and task-based mentorship.
 - Project Completion Certificate: Proof of your hands-on implementation and problem-solving skills in real-world industry projects.
 - Training Completion Certificate: Issued upon mastering our industry-aligned curriculum and passing skill-based assessments.
 - Certificate of Excellence: A prestigious award reserved for top-performing candidates who demonstrate an advanced understanding.
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Why This Journey Matters

- ▶ **Portfolio-Ready** Every CRISPR strategy you architect, every viral vector you design, and every metabolic pathway you optimize at Edlanx serves as a high-fidelity demonstration of your ability to rewrite the code of life with surgical precision and industrial scalability.
- ▶ **Global Network** Gain exclusive access to the Edlanx alumni community and industry mentors leading R&D teams at global biopharma giants, agricultural tech pioneers, and synthetic biology startups.
- ▶ **Career Excellence** We move beyond basic wet-lab techniques to provide the deep regulatory logic and genomic governance needed to solve global crises — from engineering climate-resilient crops to architecting curative gene therapies for rare genetic disorders.

This course is a living ecosystem. We constantly update our curriculum to reflect the latest shifts in Base and Prime editing, the evolution of non-viral delivery systems like LNPs, and the rapid integration of AI-driven protein folding. Your success is our shared mission.

DIRECT MENTORSHIP



Engineering biological systems and managing genetic information requires absolute molecular precision and ethical integrity. Whether you are navigating a complex off-target analysis, troubleshooting a low-titer viral production, or interpreting the regulatory requirements for a new GMO, our senior Geneticists and Bio-Process Engineers provide the direct, technical feedback needed to ensure your designs are safe, effective, and industrially ready.

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